

Section 1.1 – Definition of Common Terms

1. Which one of the following best defines agriculture?

- (a) Rearing of animals without any crop cultivation
- (b) The deliberate cultivation of crops and the rearing of animals
- (c) The study of soil and water only
- (d) Gathering wild plant products from forests

3. Shifting cultivation is a form of agriculture in which:

- (a) land is cultivated continuously for many years
- (b) the same plot is used for many generations
- (c) an area is cleared, cultivated for a few years, and then abandoned for a new area
- (d) crops are rotated on the same plot every season

5. Subsistence farming is best described as:

- (a) producing a large surplus of crops for export
- (b) cultivation carried out entirely with machines
- (c) growing crops only for animal feed
- (d) producing crops and livestock sufficient only for one's own use, without surplus for trade

7. Which statement about shifting cultivation is TRUE?

- (a) it is more advanced than settled agriculture
- (b) it makes wide use of scientific knowledge
- (c) it clears trees and cultivates the land for just one or two seasons
- (d) it requires optimized management of water

9. Tillage is best described as the:

- (a) process of cutting and gathering mature crops
- (b) removal of unwanted plants from the field
- (c) preparation of land for crop cultivation, mainly through plowing and leveling
- (d) separation of grain from the harvested stalks

11. Threshing is the process of:

- (a) cutting the mature crop in the field
- (b) removing weeds from the crop
- (c) separating the grains from the harvested plant
- (d) watering the young crop

2. Before the introduction of agriculture, societies survived mainly by:

- (a) hunting animals and gathering plants for food
- (b) trading grain with neighbouring communities
- (c) herding domesticated cattle only
- (d) cultivating the same plot for generations

4. The practice of using the same plot of land continuously over a long period to grow crops or rear livestock is called:

- (a) settled agriculture
- (b) shifting cultivation
- (c) subsistence farming
- (d) nomadic herding

6. Which of the following is a crop management practice associated with settled agriculture?

- (a) abandoning the land after every single harvest
- (b) using manure, fertilizer, and soil bunds for erosion control
- (c) clearing a fresh forest area each season
- (d) avoiding all forms of water management

8. Even today, hunter-gatherer groups still exist mainly in:

- (a) highly industrialized areas
- (b) regions with large commercial farms
- (c) cities with markets for grains
- (d) desert areas, the Arctic region, and pockets of tropical rain forest

10. In crop production, sowing refers to:

- (a) watering the growing crop
- (b) placing seeds in or on the soil to start growth
- (c) cutting the mature crop
- (d) storing the harvested seed in granaries

12. Germination is best described as the process by which:

- (a) a seed begins to grow and develop into a seedling
- (b) a mature plant produces flowers
- (c) the field is cleared before planting
- (d) grains are dried for storage

Section 1.2 – The Origin of Domesticated Crops

13. According to the text, hunter-gatherers selected plants as food sources partly because of features such as:

(a) ease of harvest, wide availability, and ease of transport (b) difficulty of harvesting and high toxicity (c) absence of leaves and large thorns (d) extreme rarity in the wild

15. Legumes used as food by hunter-gatherers are known to have been toxic, which suggests that those societies:

(a) never ate legumes at all (b) detoxified them through trial and error (c) fed them only to draft animals (d) used them exclusively as arrowhead poisons

17. According to the text, which plants were used by hunter-gatherers as sources of oil?

(a) coconut, oil palm, and olive (b) cotton, flax, and jute (c) maize, teff, and barley (d) alfalfa, clover, and Timothy

19. Which group of crops includes crops for which Ethiopia is recognized as a center of origin?

(a) wheat, rice, and maize (b) soybean, cotton, and jute (c) coconut, oil palm, and olive (d) teff, coffee, enset, and sorghum

21. The change from a hunter-gatherer culture to agriculture:

(a) took place within a single generation (b) took place gradually over thousands of years (c) occurred suddenly after the invention of the plough (d) happened at about the same time on all continents

Section 1.3 – Status of Crop Production

23. Crop production is mainly determined by:

(a) the crop type and the environment in which it is grown (b) only the amount of machinery available (c) only the number of farmers (d) only the size of the market

25. Fallowing is the practice of:

(a) planting two or more crops at the same time (b) leaving a plot of land free of crop growth for two or three seasons so the soil can recover (c) rotating

14. Wild plant species consumed by hunter-gatherers include:

(a) only domesticated rice and barley (b) only wild fruit trees (c) wild rice, grass species, wild oats, and legumes (d) only root and tuber crops

16. Root crops were widely used as food by hunter-gatherers particularly in:

(a) the Arctic region (b) desert areas (c) the temperate highlands (d) the tropical region

18. According to Vavilov, the regions where domesticated crop plants originated are called:

(a) agricultural eras (b) centers of origin (c) fallow belts (d) transition zones

20. Which crop is native to Ethiopia, where it is used as a staple food?

(a) rice (b) maize (c) teff (d) sugar cane

22. According to the text, one factor that may have influenced the development of agriculture is:

(a) religion, because particular animals and plants were kept for religious practices (b) the invention of irrigation canals (c) the spread of written language (d) the introduction of money as a means of trade

24. Technological advancement in agriculture resulted in:

(a) lower yields and smaller farms (b) the disappearance of all crop pests (c) access to improved seeds, better production practices, and better harvesting and storage facilities (d) a reduction in both plant and animal production

26. Efficient technologies that characterize modern agriculture include:

(a) stone tools and shifting cultivation (b) hunting and gathering (c) slash-and-burn clearing (d)

livestock on cropland (d) harvesting the same field twice a year

27. In correct chronological order, agriculture in industrialized regions is classified into:

- (a) Mechanical Era, Chemical Era, and Biotechnology/Information Technology Era (b) Chemical Era, Mechanical Era, and Biotechnology Era (c) Biotechnology Era, Chemical Era, and Mechanical Era (d) Mechanical Era, Biotechnology Era, and Chemical Era

29. Of Ethiopia's arable land, only about what percentage is cultivated, mainly by smallholders?

- (a) 20% (b) 40% (c) 75% (d) 90%

31. The major grain crops of Ethiopia supply approximately what percentage of the calories consumed in the country?

- (a) 25% (b) 64% (c) 80% (d) 95%

33. Which of the following is listed in the text as a principal factor responsible for low crop yields in Ethiopia?

- (a) very high levels of irrigation (b) heavy use of improved seeds and fertilizers (c) soil degradation and traditional farming systems with little technology use (d) excessive use of pesticides and machinery

Section 1.4 – Classification of Crop Plants

35. According to the text, crop plants can be classified into four major categories known as:

- (a) cereals, pulses, oil crops, and forage crops (b) annual, biennial, perennial, and seasonal crops (c) C3, C4, CAM, and aquatic plants (d) food crops, oil crops, fiber crops, and sugar crops

37. In the Ethiopian context, in terms of area of cultivation, the top five major crops are:

- (a) rice, millet, oats, rye, and triticale (b) coffee, enset, teff, flax, and sesame (c) teff, wheat, maize, sorghum, and barley (d) sugarcane, cotton, soybean, groundnut, and sunflower

39. Dicot crops such as legumes and oil seeds store their food in the:

machinery, agrochemicals, irrigation, biotechnology, and plant breeding

28. According to the text, Ethiopia has about how many million hectares of arable land?

- (a) 5 (b) 15 (c) 51 (d) 100

30. Approximately what proportion of Ethiopia's cultivated area is occupied by the five major crops (teff, wheat, maize, sorghum, and barley)?

- (a) 50% (b) 64% (c) 70% (d) 75%

32. Between 1961 and 2019, the increase in Ethiopian cereal production was associated mainly with:

- (a) the expansion of the cultivated area (b) a proportional increase in yield per unit area (c) a large rise in fertilizer use per hectare (d) the introduction of irrigation on most farms

34. According to the text, achieving food security requires:

- (a) decreasing yields and increasing their variability (b) reducing the cultivated area (c) abandoning crop rotation (d) an increase in yield and a decrease in yield variability

36. Globally, the three most important food crops in terms of volume of production are:

- (a) teff, wheat, and barley (b) wheat, rice, and maize (c) maize, sorghum, and millet (d) rice, barley, and oats

38. Monocot crops store their food primarily in the:

- (a) endosperm (b) cotyledons (c) seed coat (d) pericarp

40. Which group of plants lists major oil seed crops mentioned in the text?

(a) endosperm (b) cotyledons (c) husks (d) roots

41. Plant-derived oils supply two essential fatty acids, namely:

(a) citric and malic acids (b) acetic and formic acids
(c) linoleic and alpha-linolenic acids (d) ascorbic and folic acids

43. Which crop provides the most important fibre used for cloth-making?

(a) jute (b) cotton (c) alfalfa (d) maize

45. In the C₃ photosynthetic pathway, the first stable compound formed from the assimilation of carbon dioxide contains:

(a) two carbon atoms (b) four carbon atoms (c) three carbon atoms (d) six carbon atoms

Section 1.5 – Cropping Systems

47. A cropping system refers to:

(a) only the number of crops grown in a season (b) the types and sequences of crops and the different practices used to grow them (c) the market price of farm produce (d) the type of structure used to store grain

49. Which of the following is a disadvantage of mono cropping?

(a) the risk of total crop failure caused by pests and diseases (b) the use of uniform management practices (c) increased efficiency of planting and harvesting (d) improved soil fertility year after year

51. Mixed cropping is the practice of:

(a) growing one crop over successive years (b) growing two or more crops simultaneously on the same piece of land (c) leaving the land fallow every second season (d) cultivating a fresh area each season

53. Intercropping is the planting of more than one crop:

(a) in regular rows on the same field at the same time (b) in succession over different years (c) only after the first crop is harvested (d) on separate plots far apart

55. Which of the following is an objective of crop rotation?

(a) cotton, jute, and hemp (b) teff, wheat, and barley (c) alfalfa, clover, and Timothy (d) olive, linseed, sesame, and sunflower

42. Triglycerides, the main components of oils and fats, are composed of:

(a) glycerine and three fatty acids (b) a single fatty acid and a sugar (c) three sugar units and a protein (d) two amino acids and one fatty acid

44. Which of the following is a forage crop mentioned in the text?

(a) flax (b) sesame (c) castor (d) elephant grass

46. Which one of the following is an example of a C₄ crop given in the text?

(a) maize (b) wheat (c) teff (d) soybean

48. Mono cropping is the practice of:

(a) growing two or more crops simultaneously on the same land (b) planting a second crop after the first has grown alone for some time (c) growing a single crop type on a given plot of land (d) alternating crops of different species each season

50. Which of the following is an advantage of mono cropping?

(a) it reduces the risk of total crop failure (b) it always improves soil fertility (c) it breaks the cycle of crop pests and diseases (d) it enables uniform management and increases the efficiency of planting and harvesting

52. Which of the following is an advantage of mixed cropping?

(a) soil nutrients are used less efficiently (b) pests and diseases spread more rapidly (c) it reduces the risk of total yield loss (d) the farmer depends on the yield of only one crop

54. Relay planting is where:

(a) the same crop is planted every year on the same land (b) all crops are planted at the same time in the same rows (c) the land is left fallow for three seasons (d) a second crop is planted after the first crop has grown alone for a certain period

56. Compared with mono cropping, relay and intercropping systems:

(a) to keep the soil planted with the same species forever (b) to improve soil fertility, especially if legumes are included in the rotation (c) to reduce the number of crop species grown (d) to prevent the control of pests and diseases

(a) provide better soil cover and reduce soil erosion (b) cause crop diseases to spread much faster (c) leave the soil bare between rows (d) make efficient harvesting impossible

Section 1.6 – Indigenous Knowledge in Crop Production

57. Indigenous knowledge is best defined as:

(a) knowledge imported from foreign countries (b) knowledge obtained only in laboratories (c) traditional or local knowledge that communities have accumulated over generations in a particular location (d) knowledge based exclusively on modern machinery

59. Which society is known in the text for its traditional skills in terracing to protect against soil erosion?

(a) the Gumuz (b) the Oromo (c) the Somali (d) the Konso

58. Indigenous knowledge often develops from agricultural production problems such as:

(a) land degradation, soil erosion, and changes to the climate (b) the abundance of modern inputs (c) the exclusive use of improved seeds (d) the complete absence of farming

60. According to the text, highland farmers use kitchen ash to:

(a) control destructive insects (b) fertilize the soil and reduce soil acidity (c) protect the crop from porcupines (d) increase the smoke from their fires